JIHO KIM

C++ SYSTEMS / AI BACKENDS / UNITY INTERACTION / HAPTICS & XR

rrop@khu.ac.kr | Portfolio | GitHub

PROFESSIONAL SUMMARY

Systems-focused developer working across C++ middleware, AI backends, Unity interactions, and haptics/XR research. Built and operated the complete AI backend for a VR historical NPC on Amazon EC2 g5.xlarge, including client communication and logging. Implemented Unity game logic, interactions, objects, and animations for a LiDAR SDK-driven multi-projection CAVE escape room. Since June 2022, independently developing cross-platform C++20 audio middleware and native Godot integrations. Currently researching sensor-to-rendering MR haptic workflows at Kyung Hee University.

TECHNICAL SCOPE

AI / Cloud

Python, speech recognition, LLM integration, RAG pipeline configuration, speech synthesis, AWS, HTTP client-server communication, dialogue logging

Interactive / XR

Unity, CAVE interactions, LiDAR SDK integration, game logic, objects and animations; Godot 4 / GDExtension, OpenXR, Meta Quest 3, vibrotactile haptics

Systems / DSP

C++20, MSVC, Clang, CMake, Conan, CI/CD; FFT/STFT, Faust, miniaudio, OCCA/OKL, CUDA, OpenCL, OpenMP

Data / Transport

Cap'n Proto, CBOR, SQLite, RocksDB, TCP, UDP, WebSocket, shared-memory IPC; Windows, Linux, macOS

SELECTED AI & INTERACTIVE PROJECTS

Greatman - VR Historical NPC

AI Backend Developer / AWS Deployment

- Owned the full server and AI backend configuration, connecting speech recognition, retrieval, LLM response generation, and speech synthesis.
- Deployed and operated the backend on Amazon EC2 g5.xlarge; implemented VR client communication and dialogue logging; assisted with user experiments.
- Collaborated with a separate Unity VR client developer. Project-level field use: 42 participants and 126 sessions; related coauthor paper published in Applied Sciences (2026).

CAVE Escape Room

Unity Interaction / Game Logic / Objects / Animation

- Implemented interactions, game logic, game objects, and animations for an escape-room experience in a multi-projection CAVE space.
- Used a LiDAR SDK for data processing and integrated wall-touch input into Unity content and game behavior.

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RESEARCH & SYSTEMS EXPERIENCE

M.S. Researcher - Haptics / XR

Haptics Lab, Kyung Hee University | Mar 2025 - Present

- Developing an end-to-end Meta Quest 3 workflow for haptic texture acquisition, modeling, and vibrotactile rendering.
- Implemented 1 kHz acceleration/force sensing, AR/LSF models and interpolation, native C/C++ Godot integration, and OpenXR floating-point haptic PCM.

Project DJ Engine - Solo Open-Source Developer

Cross-Platform C++20 Audio Middleware | Jun 2022 - Present

- Designed playback, editing, DSP, device input, rhythm-game judgment, IPC and persistence across core, Godot wrapper and prebuilt distribution repositories.
- Maintained CMake/Conan builds and CI/CD delivery for Windows, Linux and macOS, including prebuilt Godot GDExtension packages.

Undergraduate Researcher - CVLab

Hoseo University | Mar 2023 - Dec 2024

- Implemented C++/OpenCV landmark-based face alignment and inference preprocessing; worked with TensorRT enqueueV3, pinned host memory and CUDA stream synchronization.

OCCAstft - GPGPU / DSP Developer

Independent Open-Source Project | Jul 2023 - Oct 2024

- Implemented portable FFT/STFT kernels using OCCA/OKL with CUDA and OpenCL execution paths, asynchronous transport and runtime fallback; profiled on NVIDIA H100.

EDUCATION

Kyung Hee University

M.S. in Computer Engineering | Mar 2025 - Present

Hoseo University

B.S. in Computer Engineering | Mar 2021 - Feb 2025

PUBLICATIONS & CONTRIBUTIONS

VR Historical NPC - Applied Sciences (2026)

Y. Kim, J. Kim, G. Ryu, T. Woo. Design and Field Validation of a VR-Based RAG Historical NPC: Knowledge Grounding and Persona Grounding Can Fail Independently. 16(21), 10707.

doi:10.3390/app162110707

Haptics / GPGPU Research

End-to-End MR haptic texture workflow, AsiaHaptics 2026 (forthcoming); related OCCAstft research published in APJCRI.

Open-Source Contributions

Merged C++ compatibility fix in Meta's RocksDB; Godot AI Pi Coding Agent client integration.